HYPERSENSITIVITY AND RELATED DISORDER

Lecturer
Nishtiman Mohammed
Internist
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OUTLINE

Hypersensitivity

Types of hypersensitivity reactions

Allergy and anaphylaxis

Investigations

Management

WHAT IS HYPERSENSITIVITY?

Hypersensitivity describes an abnormal or pathologic immune reaction that is caused by an immune response to repeated exposure to an antigen. Hypersensitivity is a set of undesirable reactions produced by the normal immune system, including allergies and autoimmunity. These reactions may be damaging, uncomfortable, or occasionally fatal.

- Hypersensitivity diseases include autoimmune diseases, in which immune responses are directed against self-antigens.
- Because these reactions tend to occur against antigens that cannot be escaped (i.e. self-antigens), hypersensitivity diseases tend to manifest as chronic problems.
- They are classified in four groups after the proposal of Gell and Coombs in 1963.

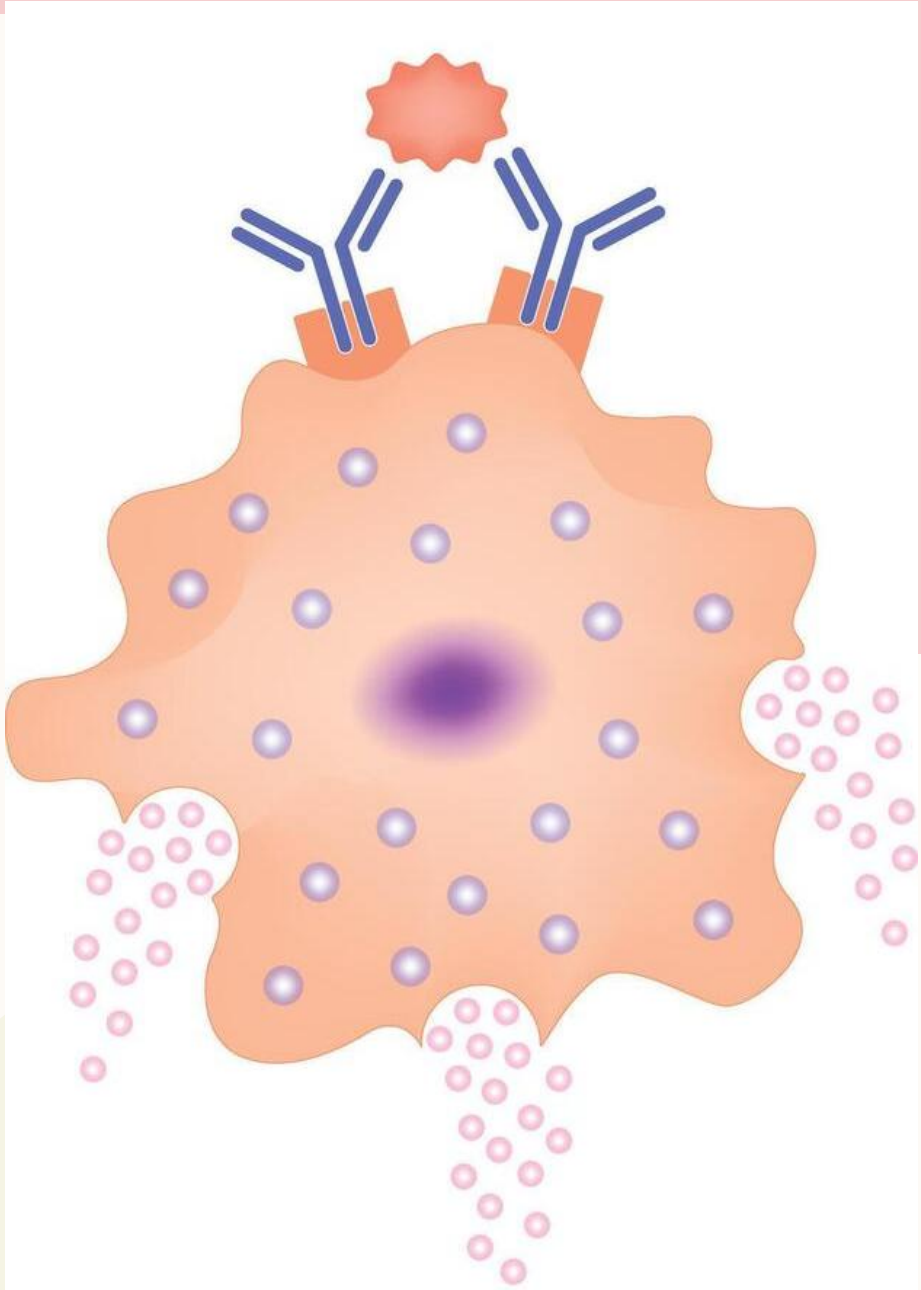


TYPES OF HYPERSENSITIVITY REACTION

GELL AND COOMBS CLASSIFICATION

Type I hypersensitivity is relevant in allergy but is not associated with autoimmune disease.

Type II hypersensitivity causes injury to a single tissue or organ and is mediated by specific autoantibodies.

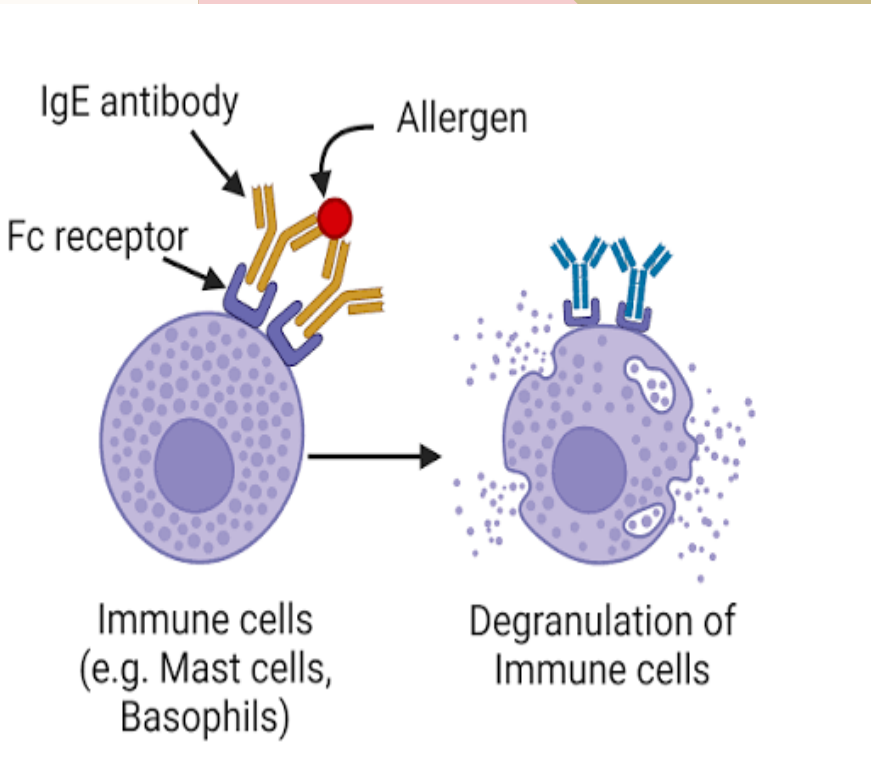


GELL AND COOMBS CLASSIFICATION

- **Type III hypersensitivity** results from deposition of immune complexes, which initiates activation of the classical complement cascade, as well as recruitment and activation of phagocytes and CD4+ lymphocytes.
- The site of immune complex deposition is determined by the relative amount of antibody, size of the immune complexes, nature of the antigen and local hemodynamics.
- Generalized deposition of immune complexes gives rise to systemic diseases such as SLE.
- **Type IV hypersensitivity** is mediated by activated T cells and macrophages, which together cause tissue damage

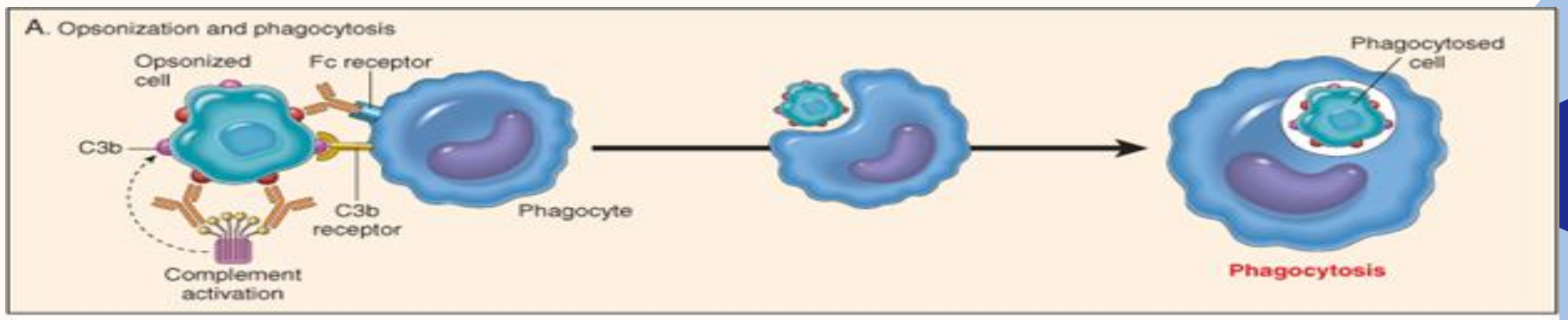
TYPE 1(IMMEDIATE) HYPERSENSITIVITY

- Also called “allergy”
- It is an IgE- mediated mast cell or basophils
- occurs in genetically susceptible atopic persons.
- occurs rapidly within minutes.
- Asthma, anaphylaxis and drug allergy belong to this group
- This type of hypersensitivity do not cause autoimmune diseases



TYPE-2(ANTIBODY-MEDIATED)HYPERSENSITIVITY

- It is an antibody (IgG or IgM)mediated
- The hypersensitivity response results from antibody binding to the surface of cell and other tissue of the host that is perceived by the immune system as foreign
- This deposition can have a variety of detrimental effects including: opsonization & phagocytosis, cell lysis, killing, and inflammation



Clinically ,this type of hypersensitivity occurs in the following situations;

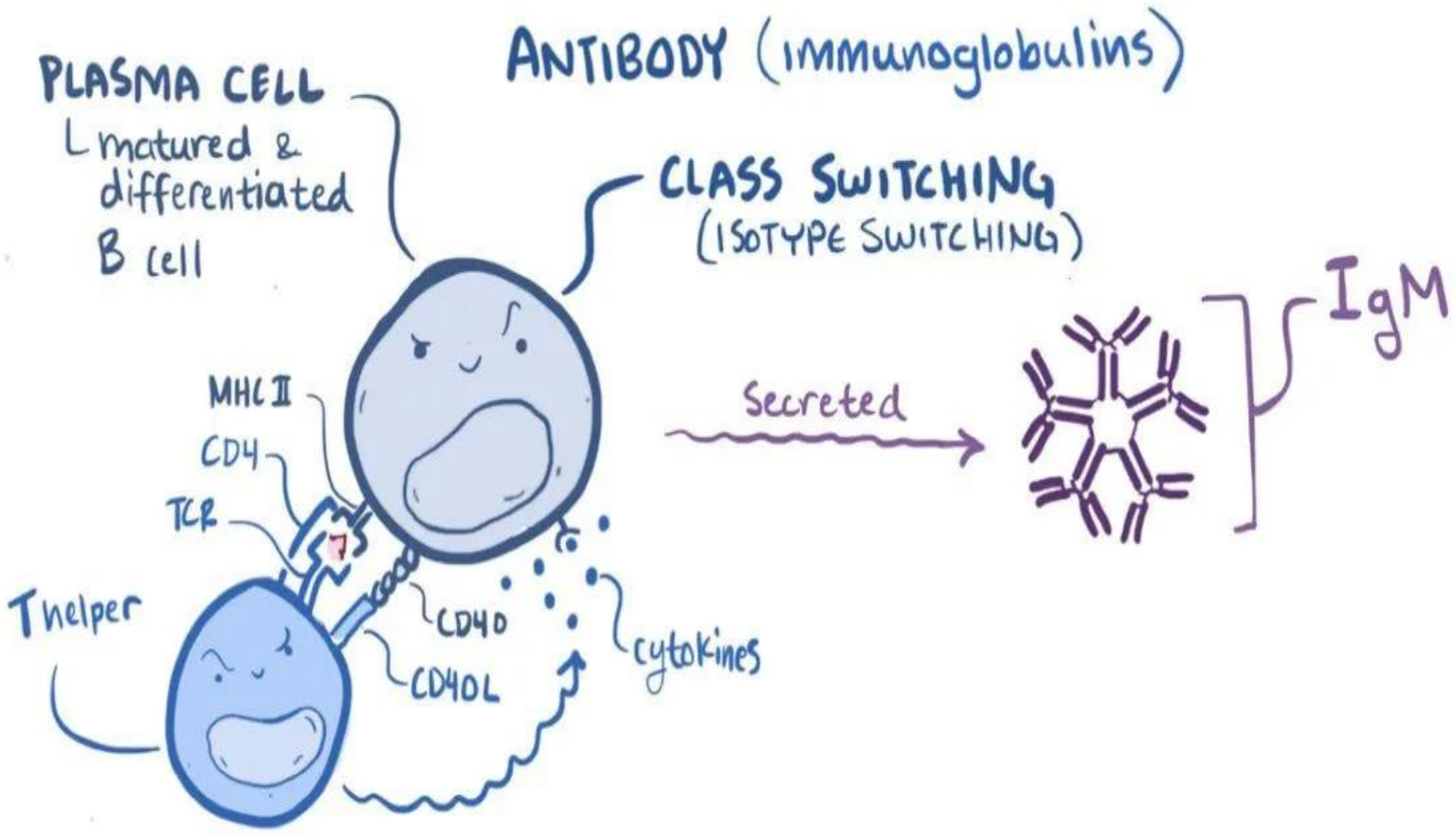
- ABO blood transfusion reaction
- Rheumatic heart disease
- autoimmune hemolytic anemia
- idiopathic thrombocytopenic purpura (ITP)

TYPE 3 (IMMUNE COMPLEX) HYPERSENSITIVITY

- Is mediated by the deposition of immune complex(Ab Ag)
- Mediated by IgG or IgM
- These pathogenic immune complexes can be formed in circulation and often deposited in the vessel walls of the joints , kidney and other tissues and cause a local inflammatory reaction.
- It can trigger classical complement pathway activation
- Examples, post-streptococcal GN, rheumatoid arthritis & SLE.

TYPE III HYPERSENSITIVITY

* mediated by IMMUNE COMPLEXES *



TYPE 4 (DELAYED TYPE (DTH) OR CELL MEDIATED) HYPERSENSITIVITY

- It is initiated by sensitized T lymphocytes(CD4+ and CD8+), phagocytes and NK cells.
- CD8+ cytotoxic T cells and CD4+ helper T cells recognize antigen in a complex with either type 1 or type 2 major histocompatibility complex
- the reaction takes two to three days to develop
- It may cause tissue injury in the skin as in contact dermatitis
- & systematically as in type 1 DM, mantoux test (tuberculosis) and chronic graft rejection.

TYPE 4 HYPERSENSITIVITY IS DIVIDED INTO 2 TYPES

1-delayed type hypersensitivity (DTH)

- CD4+ T cells secrete cytokines
- macrophages come and kill cells
- the best example is the tuberculin reaction

2- T cell mediated hypersensitivity

- Direct cell cytotoxicity.
- CD8+ T cells kill targeted cells
- This type of reaction plays an important role in graft rejection & elimination of tumor cells

ALLERGY & ANAPHYLAXIS

- ATOPY or ALLERGY is the tendency to produce an exaggerated IgE immune response to other wise harmless environmental substances(allergen).
- Affecting between 15-20% of populations
- These diseases ranging from mild to life threatening ,e.g anaphylaxis, angioedema, & allergy to insect venom.

PATHO-IMMUNOLOGY

- Initial exposure to allergen triggers the production of IgE antibodies by activated B cells.
- IgE binds to surface of mast cells & basophiles
- This leads to release of nasty mediators like histamine & prostaglandin,
- These mediators will cause vasodilatation, & smooth muscle contraction.

PREDISPOSING FACTORS

- ✓ Infection in an early life
- ✓ Family history
- ✓ Genetic predisposition
- ✓ Environmental factors such as pollutants & cigarette smoke, and the incidence of bacterial and viral infection.

INVESTIGATIONS

1- Skin prick test: is the "goal standard" of the allergy testing

2-Specific IgE test:- is an alternative to skin prick test , but less sensitive and specific

- Because IgE antibodies are unique to each allergen (for example, IgE produced in response to pollen differs from IgE produced after a bee sting), checking for specific variants in the blood can help determine if an allergy is present

3-challenge test(supervised exposure to allergen)

This include bronchial provocation testing, nasal challenge and food challenge. useful in the investigation of occupational asthma or food allergy.

4-serum level of mast cell tryptase

5-Total serum IgE level & eosinophilia:- non specific marker of atopic diseases, there is elevated level of total IgE, some times normal? , If Eosinophilia >20%, should search for non-atopic causes

ANAPHYLAXIS

- Anaphylaxis is a potentially life-threatening, systemic allergic reaction characterised by a variable combination of circulatory collapse, bronchospasm, laryngeal stridor, angioedema and urticaria.

- The risk of death is increased in patients with pre-existing asthma, particularly if this is poorly controlled, and in situations where treatment with adrenaline (epi nephrine) is delayed.

Presentation

- Anaphylaxis occurs when an allergen binds to and cross-links mem brane-bound IgE on mast cells in a susceptible individual, causing release of histamine, tryptase and other vasoactive mediators from mast cells.
- These mediators have a variety of effects, including vasodilatation, increased capillary permeability leading to hypotension, and broncho constriction.
- It can be difficult to distinguish IgE-mediated anaphylaxis clinically from non-specific degranulation of mast cells on exposure to drugs, chemicals or other triggers where IgE is not involved, previously known as anaphylactoid reactions.

CLINICAL MANIFESTATION OF ANAPHYLAXIS

- Feeling of impending doom,
- Loss of consciousness
- Conjunctival injection
- Flushing Sweating Hypotension
- Urticaria
- Wheeze, bronchoconstriction, Laryngeal obstruction Stridor
- Angioedema of lips and mucous membrane
- Abdominal pain, Diarrhoea
- Cardiac arrhythmias
- Itching of palms, soles of feet and genitalia

INVESTIGATIONS AND MANAGEMENT

- Measurement of serum mast cell tryptase concentrations is useful to confirm the diagnosis. It is important to measure tryptase in serial tests to demonstrate the temporal relationship to the triggering event.
- Specific IgE tests may be useful in confirming hypersensitivity and may be prefer able to skin-prick tests when investigating patients with a history of ana phylaxis.

- **The cornerstone of management of a severe case is the early administration of adrenaline (epinephrine),** which supports the cardiovas cular system, reduces bronchospasm and has some disease-modifying effects.
- Secondary management depends upon which organs are most affected.
- The emergency management of anaphylaxis is summarized in Box 10.7.
- If present, angioedema of the orofacial area requires careful observation and timely intervention, possibly with endotracheal intubation.
- Individuals who have recovered from an anaphylactic event should be referred for specialist assessment.
- The aims are to identify the trig ger, educate the patient regarding avoidance and management of sub sequent episodes, and to establish whether specific treatment, such as immunotherapy, is indicated.
- If the trigger factor cannot be identified or avoided, recurrence is common. Patients who have previously experienced an anaphylactic event should be prescribed self-injectable

- adrenaline (epinephrine) and they and their families or carers should be instructed in its use (Box 10.8). The use of a MedicAlert (or similar) bracelet will increase the likelihood of the injector being administered in an emergency. There is an important role for a specialist dietitian when a food trigger is the suspected allergen. Allergy in adolescence requires additional consideration and management, as set out in Box 10.9.

10.6 Differential diagnosis of anaphylaxis

Causes of hypotension

- Vasovagal syncope
- Cardiac arrhythmia
- Cardiogenic shock

Causes of respiratory distress

- Status asthmaticus
- Pulmonary embolus

Causes of laryngeal obstruction

- C1 inhibitor deficiency
- Idiopathic angioedema

Causes of generalised flushing

- Systemic mastocytosis
- Carcinoid syndrome

10.8 How to prescribe self-injectable adrenaline (epinephrine)

Prescription (normally initiated by an immunologist or allergist)

- Specify the brand of autoinjector, as they have different triggering mechanisms
- Prescribe two devices

Indications

- Anaphylaxis to allergens that are difficult to avoid:
  - Insect venom
  - Foods
- Idiopathic anaphylactic reactions
- History of severe localised reactions with high risk of future anaphylaxis:
  - Reaction to trace allergen
  - Likely repeated exposure to allergen
- History of severe localised reactions with high risk of adverse outcome:
  - Poorly controlled asthma
  - Lack of access to emergency care

Patient and family education

- Know when and how to use the device
- Carry the device at all times
- Seek medical assistance immediately after use
- Wear an alert bracelet or necklace
- Include the school in education for young patients

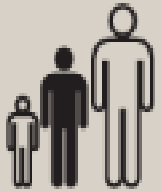
Other considerations

- Caution with β-adrenoceptor antagonists (β-blockers) in patients with anaphylaxis as they may increase the severity of an anaphylactic reaction and reduce the response to adrenaline (epinephrine)

10.7 Emergency management of anaphylaxis in adults

Treatment	Comment
Prevent further contact with allergen	Prevents ongoing mast cell activation
Ensure airway patency	Prevents hypoxia
Administer adrenaline (epinephrine) promptly: 0.5 mL 1 : 1000 solution IM (0.5 mg)	Intramuscular route important because of peripheral vasoconstriction Acts within minutes Increases blood pressure Reverses bronchospasm
If initial response is inadequate after 5 minutes, administer: Further IM bolus of 0.5 mg adrenaline (epinephrine) 500–1000 mL IV crystalloid	 Restores plasma volume
Provide supportive treatment: Nebulised adrenaline (epinephrine) (e.g. 3 mg) Nebulised β_2-agonists (e.g. salbutamol 5 mg) Oxygen	 Reduces airway oedema Reverses bronchospasm Reverses hypoxia

In refractory cases, early expert help and critical care support are vital. IM adrenaline (epinephrine) should be given every 5 minutes; a low dose IV infusion may be needed. Further boluses of IV crystalloid may also be needed.
(IM = intramuscular; IV = intravenous)



10.9 Allergy in adolescence

- **Resolution of childhood allergy**: most children affected by allergy to milk, egg, soybean or wheat will grow out of their food allergies by adolescence, but allergies to peanuts, tree nuts, fish and shellfish are frequently life-long.
- **Risk-taking behaviour and fatal anaphylaxis**: serious allergy is increasingly common in adolescents and this is the highest risk group for fatal, food-induced anaphylaxis. This is associated with increased risk-taking behaviour, and food-allergic adolescents are more likely than adults to eat unsafe foods, deny reaction symptoms and delay emergency treatment.
- **Emotional impact of food allergies**: some adolescents may neglect to carry a prescribed adrenaline (epinephrine) autoinjector because of the associated nuisance and/or stigma. Surveys of food-allergic teenagers reveal that many take risks because they feel socially isolated by their allergy.

REFERENCE

- Davidsons principle and practise of medicine, 25th edition